

Experiment-3: Decision Tree

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Python Program:

```
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree

uploaded = files.upload()
filename = list(uploaded.keys())[0]
data = pd.read_csv(filename)
print("Dataset:")
print(data)
encoders = {}
for col in data.columns:
    le = LabelEncoder()
    data[col] = le.fit_transform(data[col])
    encoders[col] = le
X = data.iloc[:, 1:-1]
y = data.iloc[:, -1]
model = DecisionTreeClassifier(criterion='entropy', random_state=42)
model.fit(X, y)
```

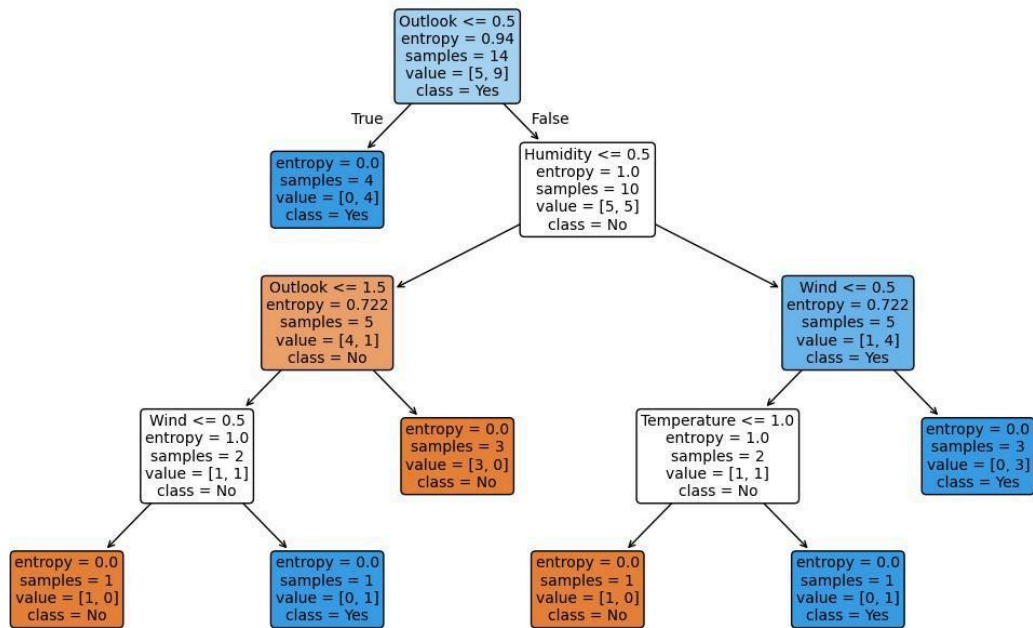
```

plt.figure(figsize=(14,8))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=encoders[data.columns[-1]].classes_,
    filled=True,
    rounded=True,
    fontsize=10
)
plt.title("Decision Tree using ID3 Algorithm")
plt.show()
new_sample = pd.DataFrame({
    "Outlook": ["Sunny"],
    "Temperature": ["Cool"],
    "Humidity": ["High"],
    "Wind": ["Weak"]
})
for col in new_sample.columns:
    new_sample[col] = encoders[col].transform(new_sample[col])
prediction = model.predict(new_sample)
result = encoders[data.columns[-1]].inverse_transform(prediction)
print("\nPrediction for New Sample:")
print(result[0])

```

Output:

Decision Tree using ID3 Algorithm



Prediction for New Sample:

No